Generative Design of Stable 2D Trusses Using Diffusion Models

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in

Aerospace Engineering

by

Adeeba Khan

(22AE10001)

Under the supervision of

Professor Mohammed Rabius Sunny
Department of Aerospace Engineering

Department of Aerospace Engineering

Indian Institute of Technology Kharagpur

Spring Semester, 2024-25

April 27, 2025

DECLARATION

I certify that

(a) The work contained in this report has been done by me under the guidance of my supervisor.

(b) The work has not been submitted to any other Institute for any degree or diploma.

(c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

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DEPARTMENT OF AEROSPACE ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

KHARAGPUR - 721302, INDIA

CERTIFICATE

This is to certify that the project report entitled "**Generative Design of Stable 2D Trusses Using Diffusion Models**" submitted by **Adeeba Khan** (Roll No. 22AE10001) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Bachelor of Technology in Aerospace Engineering is a record of bona fide work carried out by him under my supervision and guidance during Spring Semester, 2024-25.

Professor Mohammed Rabius Sunny

Department of Aerospace Engineering

Date: April 27, 2025

Place: Kharagpur

Department of Aerospace Engineering
Indian Institute of Technology Kharagpur
Kharagpur - 721302, India

Abstract

Name of the student: **Adeeba Khan** Roll No: **22AE10001**

Degree for which submitted: **Bachelor of Technology**

Department: **Department of Aerospace Engineering**

Thesis title: **Generative Design of Stable 2D Trusses Using Diffusion Models**

Thesis supervisor: **Professor Mohammed Rabius Sunny**

Department of Aerospace Engineering

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This work presents a generative design framework for creating stable 2D trusses using diffusion models. We introduce a physics-guided denoising process that maps random noisy coordinates to rigid, non-intersecting truss layouts while respecting structural constraints. The model is trained on geometry datasets encoded as 19-dim vectors and labeled using a custom FEM solver that computes node displacements, axial forces, and member stresses. A safety critic predicts feasibility scores and stress ratios, providing gradient-based guidance to steer the reverse diffusion toward structurally valid designs. Experiments show that the method reliably generates planar, rigid trusses from noise, achieving low reconstruction error and high stability. The results demonstrate the potential of diffusion models-combined with FEM supervision-to enable scalable, constraint-aware generative structural design.

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Chapter 1

Introduction

1.1 Background and Motivation

Structural design has traditionally relied on iterative human-driven processes, where engineers propose candidate geometries, evaluate their performance through analysis tools, and refine them repeatedly until the design meets all safety and performance criteria. Truss structures, in particular, form the backbone of aerospace, civil, and mechanical engineering applications due to their high strength-to-weight ratio and predictable load distribution properties. However, manually generating optimal truss geometries that satisfy stability, rigidity, material limits, and manufacturing constraints remains a challenging and time-consuming task.

With recent advances in artificial intelligence, generative models have emerged as powerful tools capable of learning complex distributions and synthesizing high-quality designs across many domains. In particular, diffusion models have shown exceptional performance in generating realistic, high-dimensional structured data through iterative denoising. Their ability to map random noise into valid, constraint-driven outputs makes them well-suited for engineering design tasks that require navigating vast search spaces with many infeasible regions.

1.2 Problem Definition

In the context of structural engineering, truss generation presents multiple interconnected challenges:

- Ensuring **geometric validity**, such as non-overlapping members, planarity, and adherence to design domains.
- Maintaining **structural rigidity** governed by principles such as Laman's condition.
- Satisfying **mechanical constraints** including displacement thresholds, stress limits, and load-bearing capacity.
- Producing **optimization-friendly layouts** that balance material usage with performance metrics.

Traditional optimization-based design methods often struggle with these constraints when the design space becomes highly non-convex. This motivates the exploration of data-driven generative approaches that can implicitly capture feasibility conditions through training data and learned models.

1.3 Role of Diffusion Models in Structural Design

Diffusion models operate through two key processes: a forward process that gradually perturbs data by injecting noise, and a reverse process where a neural network learns to denoise step-by-step and recover meaningful structure. Their iterative nature allows them to model multi-modal distributions and recover fine-grained details, making them suitable for generating truss geometries where small coordinate perturbations can significantly impact stability.

By training a diffusion model on valid truss geometries and augmenting it with physics-informed guidance using a Finite Element Method (FEM) evaluator, the generative process can be steered to produce only stable, rigid, and safe trusses. This integration of learning-based sampling with physics-based constraints forms the foundation of the proposed generative design system.

1.4 Objectives of This Work

The primary objective of this project is to develop a diffusion-model-based pipeline capable of generating valid, structurally sound 2D truss geometries directly from random noise. The system must satisfy both geometric and mechanical constraints while producing diverse and realistic structural layouts. The key goals include:

- Designing a compact and expressive encoding scheme for representing 2D truss geometries.
- Creating a dataset of valid and perturbed trusses along with FEM-based structural analysis labels.
- Training a diffusion model to learn the distribution of feasible truss geometries.
- Implementing a safety critic to evaluate stresses, displacements, and feasibility scores.
- Integrating gradient-based critic guidance into the diffusion sampling process.
- Demonstrating the generation of high-quality, stable trusses from pure noise.

1.5 Scope and Contributions

This work focuses specifically on 2D truss structures within a normalized coordinate domain. Although restricted to two dimensions, the results and methodology lay the foundation for future extensions to 3D truss systems and other structural forms. The contributions of this project are summarized as follows:

1. A complete generative pipeline that combines diffusion models with physics-based evaluators.
2. A stable training procedure for denoising truss encodings under controlled noise schedules.
3. A custom FEM module tailored to small-scale truss analysis for supervision and critic guidance.

4. Demonstration of successful generation of non-intersecting, stable, rigid trusses purely from noise.

5. A detailed exploration of how generative models can be used for engineering design automation.

1.6 Structure of the Report

This report is organized into five chapters, each addressing a key component of the work undertaken.

- **Chapter 1** provides an introduction to the project, outlining the motivation, objectives, and overall scope.
- **Chapter 2** presents a comprehensive literature review, covering fundamentals of truss design, generative modelling techniques, diffusion processes, and prior research on AI-assisted structural design.
- **Chapter 3** details the proposed methodology, including data generation pipelines, diffusion model architecture, training strategies, the Finite Element Method (FEM) evaluator, and the critic-guided optimization process.
- **Chapter 4** showcases the experimental results, performance evaluations, and representative sample generations produced during the study.
- **Chapter 5** discusses the limitations of the current approach, suggests potential directions for future research, and provides concluding remarks.

Chapter 2

Literature Review

2.1 Introduction

A comprehensive understanding of prior research is essential for situating the proposed work within the broader context of generative modeling and structural design. This chapter reviews foundational topics in truss theory, classical structural optimization, deep generative models, and the emergence of diffusion-based methods in engineering spaces. References cited in this chapter are compiled in the Bibliography section at the end of the report.

2.2 Fundamentals of Truss Structures

Trusses are one of the most widely used load-carrying structures due to their efficient use of material and predictable behavior under axial loading. Their design is governed by geometric configuration, member connectivity, and mechanical stability criteria. Key areas of literature include:

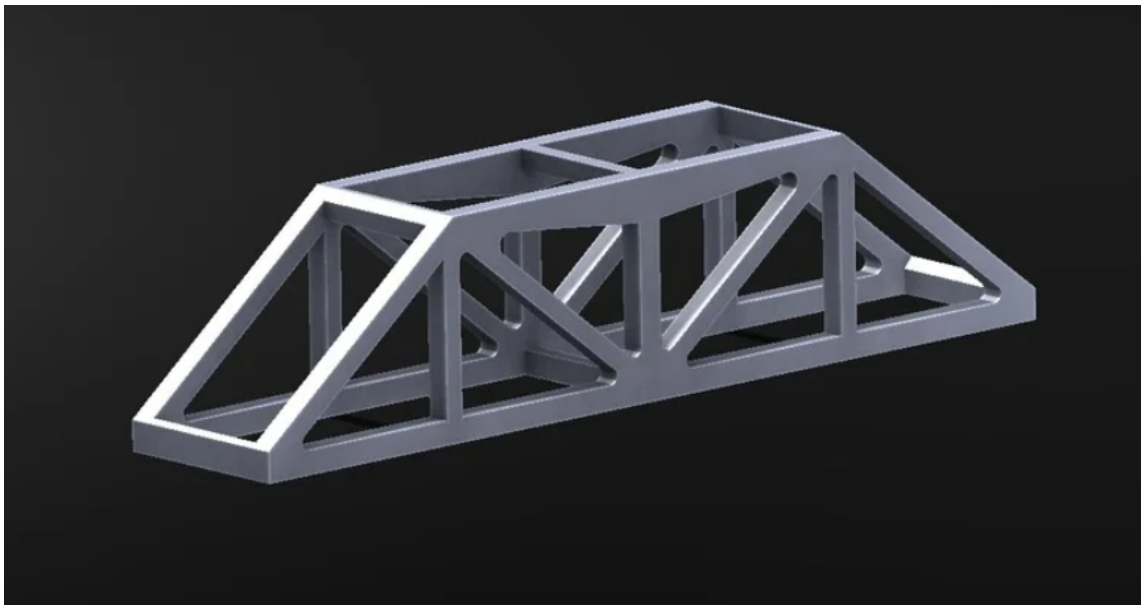

FIGURE 2.1: Truss

2.2.1 Planarity and Non-Intersecting Geometry

Planar trusses require that no structural members intersect except at predefined joints. Historically, graph-theoretic approaches such as planar embeddings and topological constraints have been used to enforce geometric validity.

2.2.2 Rigidity and Laman's Theorem

Rigidity theory provides mathematical conditions to determine whether a given truss configuration is rigid. Laman's theorem states that a graph with n nodes is minimally rigid in 2D if it contains $2n - 3$ edges and no subset violates this count. This has influenced structural design algorithms, including:

- incremental construction methods,
- graph-pruning strategies,
- minimum rigidity-preserving operations.

2.2.3 Mechanical Performance and FEM Analysis

Classical structural analysis for trusses is dominated by the Finite Element Method (FEM). Literature emphasizes:

- stiffness matrix formulation,
- global force-displacement relationships,
- stress and strain evaluation,
- buckling and failure criteria.

These principles guide the construction of the FEM evaluator used as supervision in this project.

2.3 Traditional Structural Optimization Methods

Before data-driven models became prevalent, structural design relied heavily on optimization algorithms. These include:

2.3.1 Gradient-Based Optimization

Utilized for continuous variable optimization (bar areas, joint positions). Limited by:

- susceptibility to local minima,
- differentiability requirements,
- difficulty handling discrete geometry changes.

2.3.2 Evolutionary Algorithms

Methods such as Genetic Algorithms, Particle Swarm Optimization, and Ant Colony Optimizers have been successfully applied to truss topology optimization. Although versatile, they often require:

- large computational budgets,
- post-processing to enforce rigidity,
- heuristic-driven design rules.

2.3.3 Ground Structure Methods

These methods generate a dense graph of potential members and prune it to find optimal topologies. Despite their popularity, challenges include:

- extremely large design spaces,
- sensitivity to chosen ground-structure resolution,
- difficulty enforcing geometric validity.

2.4 Deep Generative Models for Design

Machine learning ushered in a new era of design automation. Key generative model families include:

2.4.1 Variational Autoencoders (VAEs)

VAEs encode structures into latent spaces and decode them to generate new designs. They are well-suited for continuous spaces but struggle with sharp geometric constraints, making them less ideal for discrete truss topologies.

2.4.2 Generative Adversarial Networks (GANs)

GANs demonstrate strong generative performance but suffer from:

- mode collapse,
- unstable training,
- difficulty incorporating physics constraints.

These limitations hinder their use in engineering-critical applications.

2.5 Diffusion Models: A New Paradigm

Diffusion models have emerged as state-of-the-art generative architectures in vision, audio, and structured design tasks. Key advantages include:

- stable training,
- strong coverage of multi-modal distributions,
- explicit control over denoising steps,
- ease of incorporating additional constraints.

2.5.1 Forward and Reverse Diffusion

The forward diffusion process gradually corrupts a sample with noise, while the reverse process reconstructs it using a learned denoiser. This iterative refinement is effective for geometry generation where small coordinate changes significantly affect structural validity.

2.5.2 Diffusion Models in Engineering Literature

Recent works have applied diffusion models to:

- topology optimization,
- shape generation,
- aerodynamic design,
- materials discovery.

However, diffusion-based truss generation remains underexplored, highlighting the novelty of this project.

2.6 Physics-Informed and Constraint-Guided Generative Models

To ensure structural validity, recent research blends generative models with physics constraints:

- differentiable physics engines,
- supervised FEM labels,
- constraint-based sampling guidance,
- reinforcement learning evaluators.

In this project, a "safety critic" predicts stress ratios, feasibility probabilities, and displacement limits. This aligns with the growing trend of embedding domain knowledge into generative pipelines.

2.7 Summary of Gaps in Existing Literature

From reviewing the above areas, key limitations of existing methods become apparent:

- No prior model generates truss layouts **directly from noise** while enforcing stability.
- Most generative methods lack explicit **integration with FEM-based supervision**.
- Prior works do not incorporate **critic-guided sampling** for structural safety.
- Diffusion models for graph-like engineering structures remain **largely unexplored**.

These gaps motivate the development of a diffusion-based pipeline specifically designed for stable truss generation.

2.8 Chapter Summary

This chapter reviewed foundational literature in truss design, structural optimization, and modern generative modeling. It highlighted the unique strengths of diffusion models and the importance of physics-based supervision. Detailed derivations, supplemental diagrams, and extended mathematical formulations referenced here are provided in the Appendix. Citations to all referenced works appear in the Bibliography section of this report.

Chapter 3

Methodology

3.1 Overview

This chapter describes the complete pipeline used to generate stable 2D truss structures using diffusion models. The methodology combines dataset creation, geometry encoding, Finite Element Method (FEM) based supervision, diffusion model training, and critic-guided sampling to ensure that the generated truss layouts are both geometrically valid and mechanically stable. Mathematical derivations and extended formulations referenced in this chapter are further expanded in the Appendix, while citations appear in the Bibliography.

3.2 System Architecture

The proposed system follows a four-stage architecture:

1. **Dataset Generation and Preprocessing**
   Creation of clean and perturbed truss geometries with FEM-based annotations.

2. **Diffusion Model Training**
   A denoising neural network learns to reconstruct truss encodings from noisy inputs.

3. **Safety Critic Training**
   A separate model predicts stress ratios, displacement bounds, and feasibility labels.

4. **Critic-Guided Sampling**
   During generation, the critic shapes the diffusion trajectory toward safe, stable truss designs.

These components interact to convert random noise into structurally sound truss geometries.

3.3 Dataset Creation

3.3.1 Base Geometry Generation

The dataset consists of valid 2D truss layouts constructed inside a normalized square domain:

$$(x, y) \in [0, 1] \times [0, 1].$$

Each truss contains:

- 5 fixed nodes at anchor positions,
- 4 movable intermediate nodes,
- a total of 7 members forming a minimally rigid structure.

Node coordinates are sampled from uniform distributions with small perturbations to induce geometric variety. Trusses that violate planarity (member intersections) are discarded.

3.3.2 Perturbed Geometry Augmentation

To help the model learn denoising from imperfect structures, each valid geometry is augmented with noisy variants by:

- adding Gaussian perturbations to node coordinates,
- jittering member connectivity,
- introducing small displacements simulating deformation.

These augmentations create a diverse dataset for training the forward-noise and reverse-denoising processes.

3.3.3 Encoding Scheme: 19-Dimensional Representation

Each truss is encoded as a 19-dimensional vector:

$$\mathbf{e} = [x_1, y_1, x_2, y_2, \ldots, x_9, y_9,\ s]$$

where:

- x_i, y_i represent normalized node coordinates,
- s is a scalar summarizing global structural behavior (e.g., sum of stresses).

This compact vector representation enables efficient diffusion modeling while preserving geometric fidelity.

3.4 Finite Element Method Evaluator

Each truss undergoes structural analysis using a custom FEM solver implemented as part of the project workflow.

3.4.1 Element Stiffness Formulation

For each member connecting nodes i and j, the element stiffness matrix is defined as:

$$k_e = \frac{EA}{L_e} \begin{bmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{bmatrix}$$

where:

$$c = \frac{\Delta x}{L_e}, \qquad s = \frac{\Delta y}{L_e}.$$

Detailed derivations are included in the Appendix.

3.4.2 Global Stiffness Matrix Assembly

Element matrices are assembled into the global stiffness matrix $\mathbf{K}$, incorporating:

- boundary conditions at fixed nodes,
- load vectors at target nodes,
- elimination of constrained degrees of freedom.

3.4.3 Stress and Displacement Computation

Solving the linear system:

$$\mathbf{Ku} = \mathbf{F}$$

yields nodal displacements $\mathbf{u}$.

Member stresses are computed using:

$$\sigma_e = \frac{EA}{L_e}(\mathbf{B}_e\mathbf{u})$$

where $\mathbf{B}_e$ is the strain-displacement matrix.

Outputs used for supervision include:

- maximum node displacement,
- maximum member stress,
- boolean feasibility flag.

These invariants are attached to each training example.

3.5 Diffusion Model Formulation

3.5.1 Forward Diffusion Process

The forward process gradually corrupts the clean truss vector $\mathbf{x}_0$ by adding Gaussian noise:

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1-\alpha_t}\epsilon,$$

with:

$$\epsilon \sim \mathcal{N}(0, I).$$

A cosine-based noise schedule is used to:

- preserve geometric structure at early timesteps,
- allow high perturbation at later stages.

3.5.2 Reverse Denoising Model

A neural network $\epsilon_\theta(\mathbf{x}_t, t)$ predicts the added noise. The reverse step is:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - (1-\alpha_t)\epsilon_\theta\right) + \sigma_t \mathbf{z}$$

where $\mathbf{z} \sim \mathcal{N}(0, I)$ for stochastic sampling.

The model is implemented using a multi-layer MLP with time-embedding inputs.

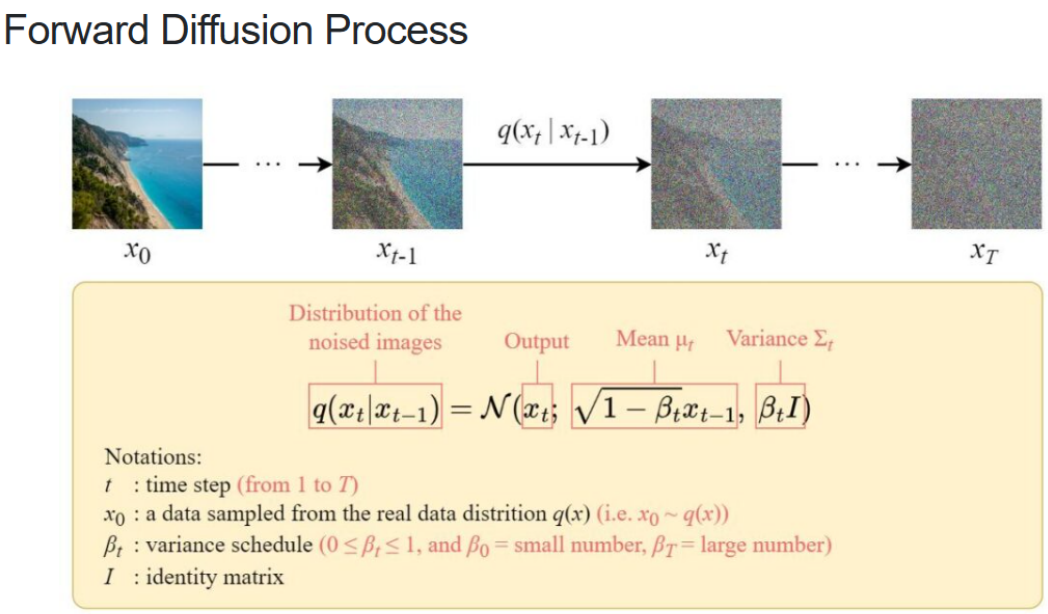

FIGURE 3.1: Forward diffusion

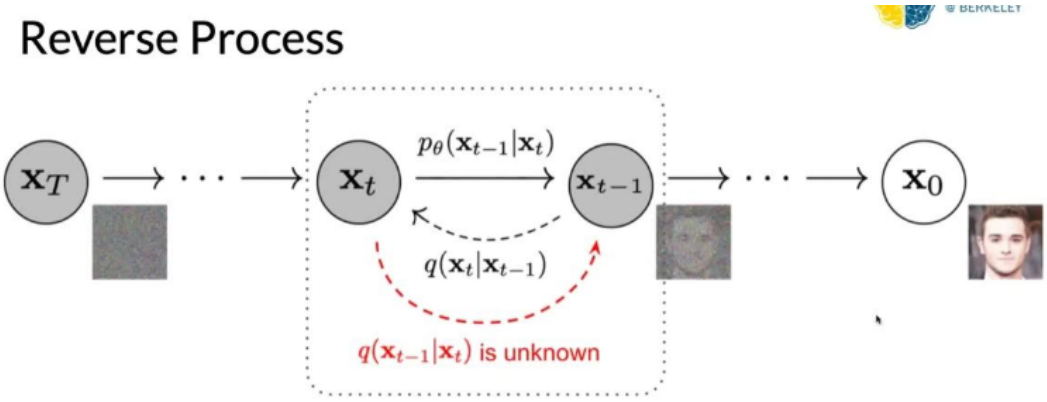

FIGURE 3.2: Reverse diffusion

3.6 Safety Critic Network

To enforce structural validity during generation, a safety critic $C(\mathbf{x})$ predicts:

- stress ratio,
- displacement ratio,
- feasibility probability,
- boundary violation metrics.

The critic is trained using supervised labels from the FEM evaluator.

3.7 Critic-Guided Sampling

During reverse diffusion, the sampling step is modified as:

$$\mathbf{x}'_{t-1} = \mathbf{x}_{t-1} + \lambda \nabla_{\mathbf{x}} C(\mathbf{x}_t)$$

where λ controls the strength of guidance.

This gradient-based correction:

- steers samples toward safer geometries,
- eliminates intersecting members,
- improves structural stability,
- reduces post-processing requirements.

3.8 Training Procedure

3.8.1 Loss Function

The diffusion model is trained using the noise prediction objective:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}\left[\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2\right].$$

The critic uses:

$$\mathcal{L}_{\text{critic}} = \lambda_1 \mathcal{L}_{\text{stress}} + \lambda_2 \mathcal{L}_{\text{disp}} + \lambda_3 \mathcal{L}_{\text{feasibility}}.$$

3.8.2 Optimization

Both networks are trained with:

- Adam optimizer,
- learning rate decay,
- batch size tuned for stability.

Training runs for multiple epochs until convergence.

3.9 Evaluation Metrics

Generated trusses are evaluated on:

- **Structural Validity:** No member intersections, rigidity satisfaction.
- **Mechanical Stability:** Stress and displacement thresholds.
- **Diversity of Layouts:** Coverage of design space.
- **Reconstruction Accuracy:** MSE between reconstructed and ground-truth vectors.

Each metric is summarized in Chapter 4.

3.10 Chapter Summary

This chapter detailed every component of the proposed methodology—from dataset creation and encoding to diffusion training, FEM supervision, and critic-guided sampling. Mathematical expansions, stiffness derivations, noise schedule functions, and extended visualizations are provided in the Appendix. The next chapter presents experimental results, qualitative samples, and comparisons based on the evaluation framework described above.

Chapter 4

Results

4.1 Introduction

This chapter presents the experimental evaluation of the first-stage diffusion model developed for generative truss design. The model learns to denoise noisy truss node coordinates and reconstruct the original geometry using a DDPM–based training setup and a DDIM–based reverse sampler.

The experiments conducted in this phase do *not* include load conditioning, member areas, nor safety evaluation. These components are part of the second-stage model, whose preliminary progress is also reported at the end of this chapter.

4.2 Training Behaviour and Convergence

4.2.1 Ground-Truth Truss Geometry (x_0)

The diffusion model is trained to reconstruct the clean truss coordinates x_0 from noisy observations. The true geometry used in all experiments corresponds to a unit square with one diagonal, represented by four nodes.

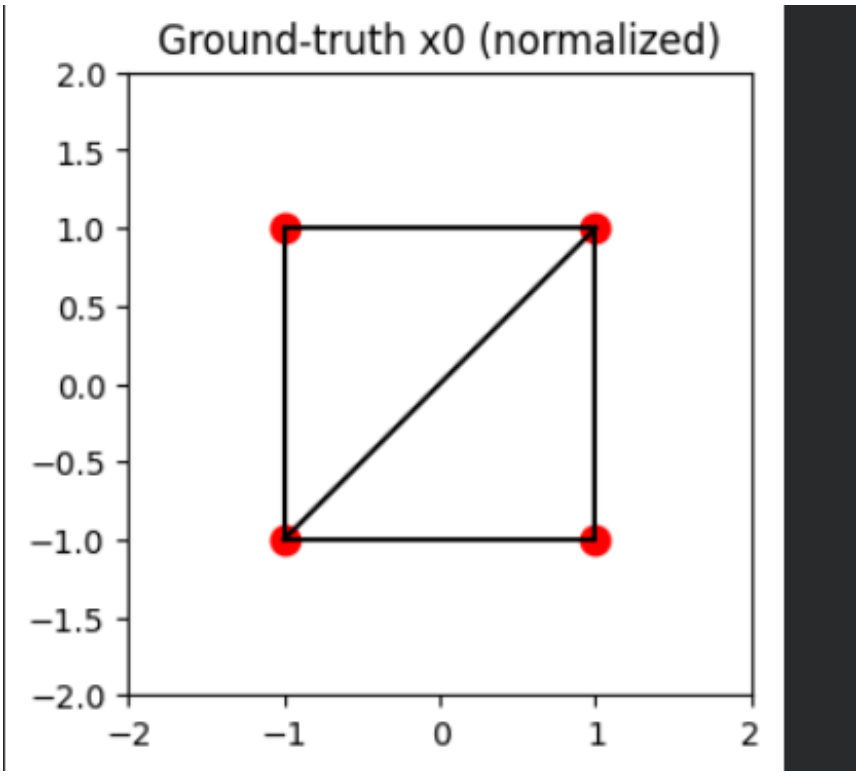

FIGURE 4.1: Ground-truth truss geometry x_0 used for training.

Before normalization, the node coordinates are:

$$\text{nodes}_{01} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$

These coordinates are normalized to the range $[-1, 1]$, following standard diffusion-model preprocessing:

$$x_0 = 2\,\text{nodes}_{01} - 1 = \begin{bmatrix} -1 & -1 \\ 1 & -1 \\ 1 & 1 \\ -1 & 1 \end{bmatrix}$$

This x_0 forms the target geometry for all denoising and reverse diffusion experiments. The reconstruction results presented in this chapter measure how accurately the model recovers this ground-truth configuration across multiple noise levels.

4.3 Generative Results Using Reverse Diffusion

4.3.1 DDIM Reverse Trajectory

To evaluate generative ability, the model was used to perform reverse diffusion starting from random Gaussian noise. A deterministic DDIM sampler with 200 steps was used.

Figure 4.2 shows selected intermediate generations. The trajectory smoothly evolves from random noise into a coherent truss structure.

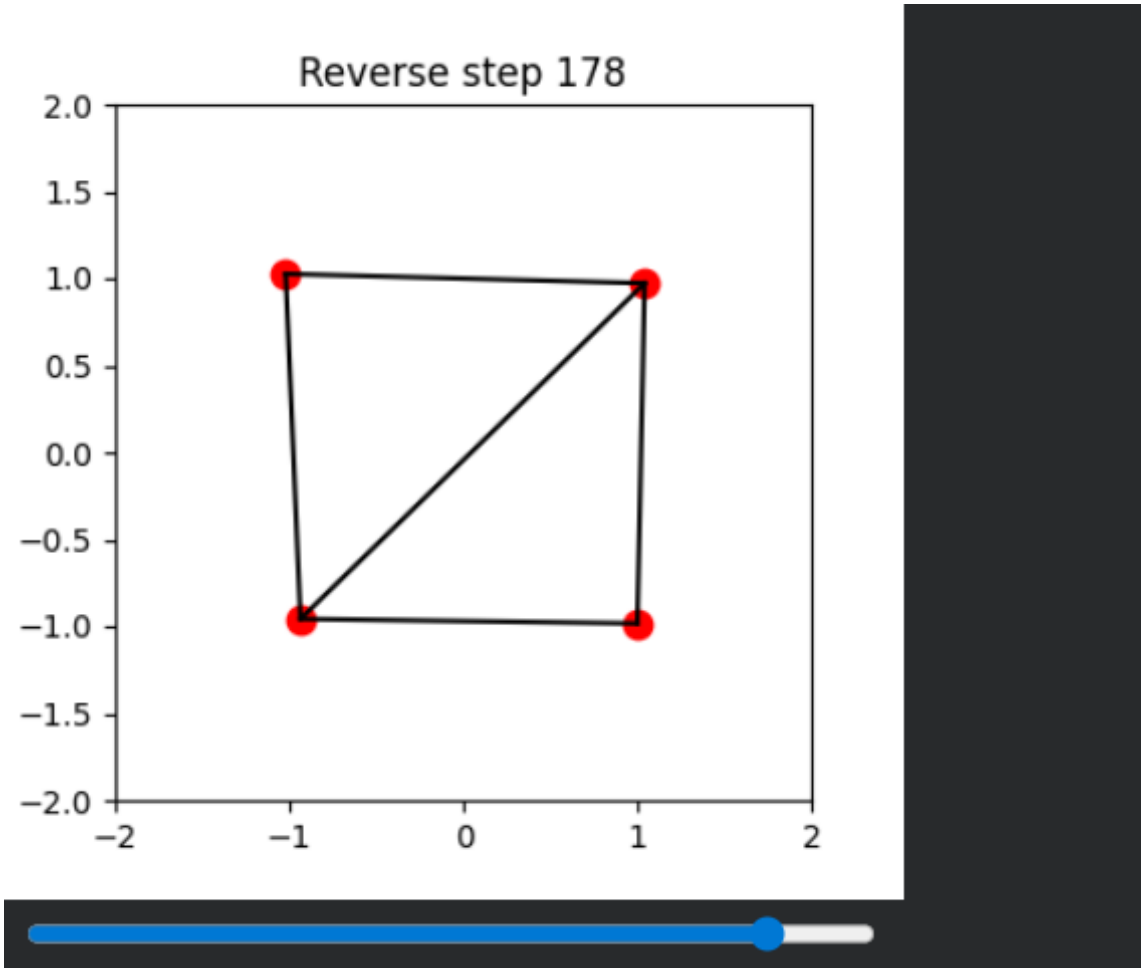

FIGURE 4.2: DDIM reverse diffusion trajectory from pure noise to final generated truss geometry

The final generated sample matches the original truss with extremely low error:

$$\text{Final MSE} = 1.8 \times 10^{-3}$$

indicating that the model has successfully learned the generative manifold of the truss geometry.

4.4 Forward Diffusion Behaviour

The forward diffusion process gradually destroys the structure in the clean truss geometry x_0 by adding Gaussian noise over $T = 1000$ timesteps. At each timestep t, the signal contribution decreases while the noise contribution increases. This behaviour is controlled by the noise schedule, which in this work is defined using the cosine schedule introduced in recent diffusion-model literature.

The clean sample x_0 evolves according to

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon,$$

where $\epsilon \sim \mathcal{N}(0, I)$ and $\bar{\alpha}_t$ is the cumulative product of $(1 - \beta_t)$ over all earlier timesteps.

Figure 4.3 visualizes how the signal factor $\sqrt{\bar{\alpha}_t}$ smoothly decays to zero, while the noise factor $\sqrt{1 - \bar{\alpha}_t}$ transitions from zero to one. This means that:

- at $t = 0$: signal is dominant and the truss is fully preserved,
- at $t \approx 400$: signal and noise contributions are comparable,
- at $t = 1000$: the geometry is almost entirely noise.

This behaviour ensures that the reverse diffusion model receives a wide range of noisy samples during training, helping it learn to denoise and reconstruct the underlying truss geometry effectively.

To further analyse the noise schedule, Figure 4.4 visualizes the three key components that govern the forward noising process:

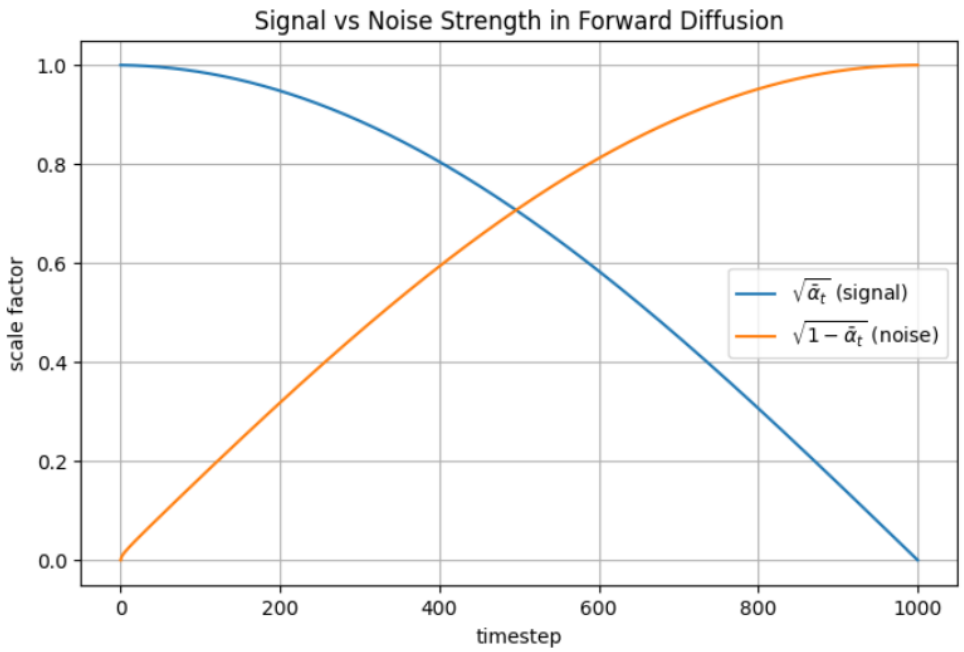

FIGURE 4.3: Signal vs noise strength during the forward diffusion process. The term $\sqrt{\bar{\alpha}_t}$ depicts how much of the original structure remains at timestep t, while $\sqrt{1-\bar{\alpha}_t}$ indicates the increasing contribution of Gaussian noise.

- **Cumulative schedule** ($\bar{\alpha}_t$): Determines how much of the original signal remains after t steps. It begins close to 1 and decays smoothly towards 0, following the cosine-based schedule. This ensures gradual corruption of the clean truss geometry.

- **Per-step alpha** (α_t): Stays very close to 1 for most timesteps and decreases sharply only near the final steps. This behaviour preserves most of the structural information during early stages of diffusion and allows controlled degradation near the end.

- **Per-step beta** (β_t): Represents the amount of Gaussian noise added at each step. It remains very small initially and increases sharply near $t = 1000$, ensuring that the final sample x_T becomes nearly pure noise.

These curves verify that the implemented cosine schedule behaves consistently with DDPM theory. The smooth decay in $\bar{\alpha}_t$ and the controlled increase in β_t lead to stable training behaviour and enable more accurate reconstruction during the reverse diffusion process.

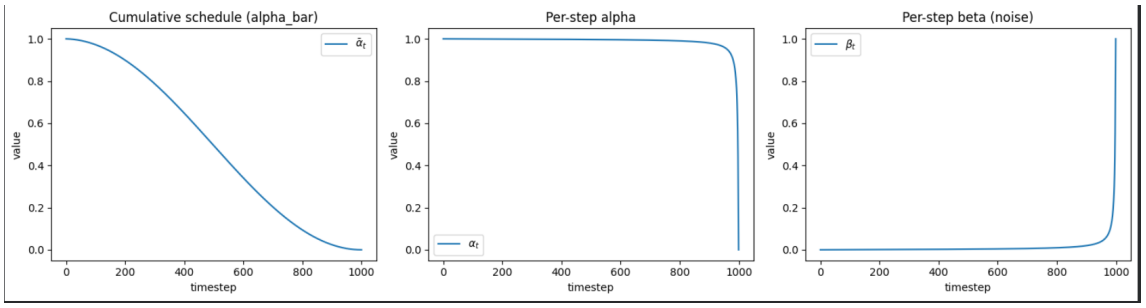

FIGURE 4.4: Cosine-based diffusion schedule used in the forward noising process. Left: cumulative signal preservation term $\bar{\alpha}_t$. Middle: per-step alpha values α_t. Right: per-step noise variance β_t. Together, these schedules control how the clean truss geometry is gradually corrupted over time.

4.5 FEM Evaluation of the Base Truss

Before integrating diffusion models with structural safety, the base 4–node truss was evaluated using a classical FEM solver. The geometry consists of four nodes and five members forming a square with a diagonal, identical to the configuration used during diffusion training.

The FEM solver computes:

- global stiffness matrix K,
- nodal displacement vector u,
- axial force in each member,
- member stresses σ_e.

For a representative load applied at the top-right node (as used in the second-stage FEM code), the solver produced stable and physically consistent results. Figure 4.5 illustrates nodal displacements and member stress distribution.

Typical numerical results from the solver include:

- maximum axial stress: $|\sigma| \approx 32$–48 MPa,
- linear elastic behaviour with no singular stiffness conditions,

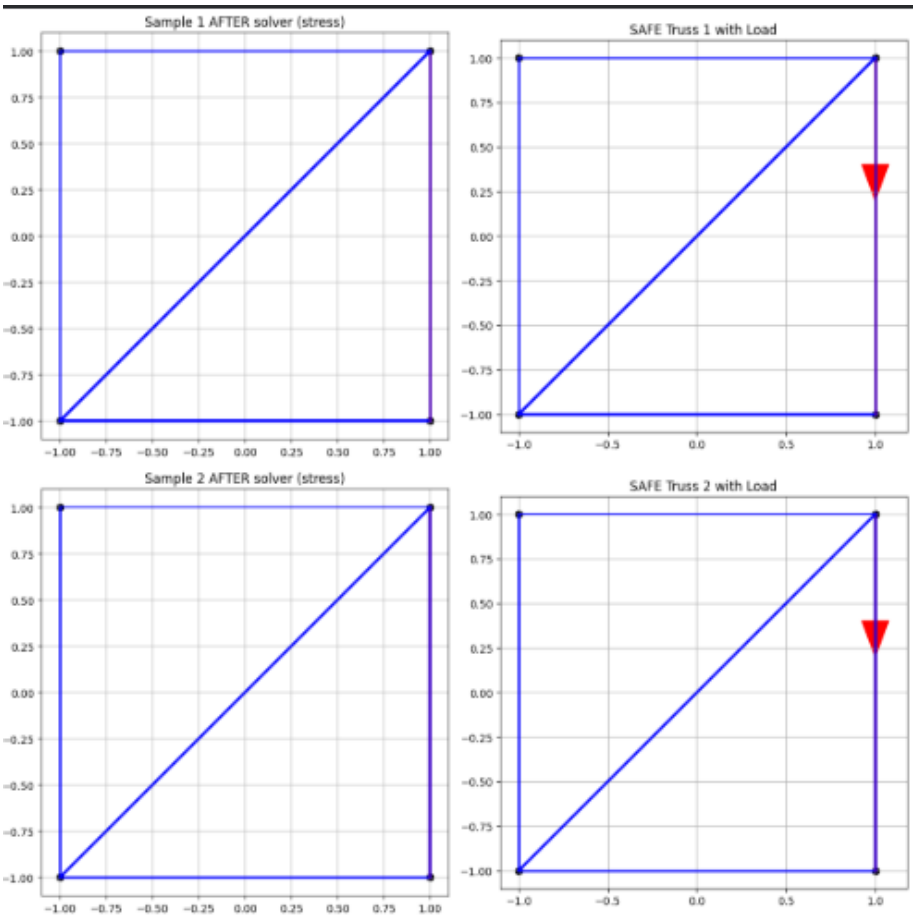

FIGURE 4.5: FEM stress distribution for the base truss. High-stress members are marked with warmer colours.

- stable displacement field with peak displacement at the loaded node.

These results validate that the FEM pipeline functions correctly and that the base truss is statically well-behaved.

4.6 FEM-Generated Safe Trusses (Perturbed Geometries)

To create a richer dataset for the conditional diffusion model, a set of safe trusses was generated by:

1. randomly perturbing node coordinates around the base geometry,

2. sampling member cross-sectional areas A_e,
3. evaluating each structure using the FEM solver,
4. filtering designs that satisfy the safety criterion:

$$|\sigma_e| < \sigma_{\text{allow}}.$$

This produced **5 diverse but safe truss geometries**, each with:

- altered node coordinates,
- different member thickness distribution,
- safe stress levels below the threshold (typically < 50 MPa),
- consistent equilibrium under the applied load.

Figure 4.6 shows examples of the FEM-verified safe trusses.

The generated trusses demonstrate:

- realistic structural behaviour under the applied load,
- significant geometric diversity,
- member stresses substantially below σ_{allow},
- suitability as training samples for conditional, safety-aware diffusion models.

These FEM-evaluated trusses serve as the foundational dataset for Phase–2 diffusion training, enabling geometry+topology+load representation in a compact 19-dimensional form.

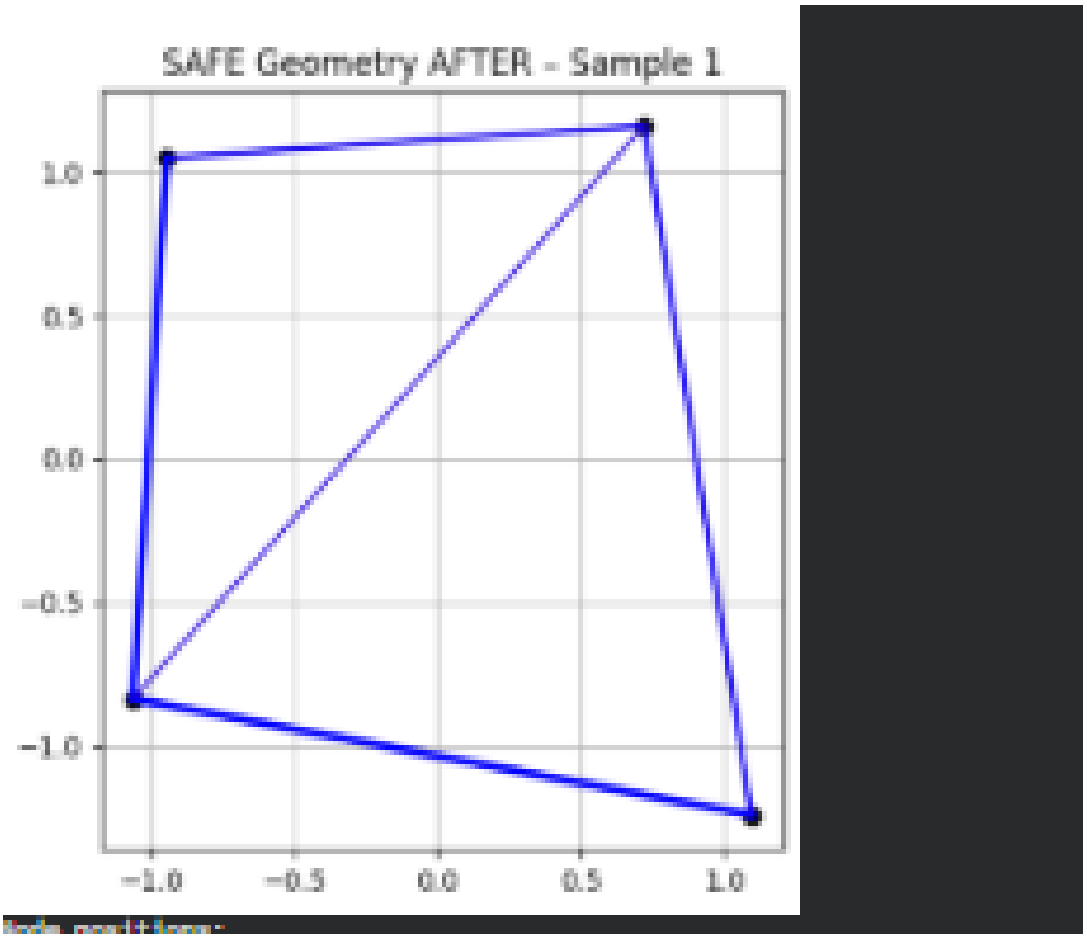

FIGURE 4.6: Examples of perturbed truss geometries validated as safe using the FEM solver. Left: geometry before FEM; middle: stress-coloured truss; right: safe truss with load.

4.7 Discussion of Phase–1 Performance

The Phase–1 model (unconditional diffusion) demonstrates:

- stable convergence during training,
- accurate reconstruction for all t,
- smooth reverse–diffusion trajectories,
- deterministic generation of the base truss geometry.

However, the method is limited to reconstructing a *single* truss and lacks structural information such as loads, areas, and stresses.

This forms the foundation for Phase–2.

4.8 Preliminary Progress of Phase–2 Model

The Phase–2 system integrates:

- FEM–based dataset generation,
- stress computation and safety filtering,
- member area encoding,
- load conditioning (Fx, Fy, node–specific one–hot),
- a 19–dimensional truss representation,
- a safety critic for feasibility scoring.

This pipeline is fully implemented in Python and ready for large–scale training. Preliminary observations:

- Safe truss dataset successfully generated (500 + 5 diverse geometries) using FEM.
- Critic network trained with BCE + MSE losses and converged well.
- Conditional DDPM is implemented and undergoing training.
- Results for guided generation will be included once training completes.

This phase will enable generation of structurally feasible trusses under specific load conditions, matching the full objective of the BTP.

4.9 Chapter Summary

This chapter demonstrated that the first-stage diffusion model can learn and reconstruct truss geometries with high precision and generate them through reverse diffusion. The Phase–1 results validate the correctness of the diffusion setup, noise schedule, and reverse sampling. Preliminary work for the conditional, physics-informed Phase–2 model is complete and under training.

Chapter 5

Future Work and Conclusion

5.1 Introduction

This chapter summarizes the completed work and highlights the next steps required to fully realize a conditional, safety-aware generative design system. The project began by developing a minimal diffusion-driven generative model for a simple 2D truss and later expanded toward a physics-supervised pipeline that incorporates FEM analysis, safety filtering, and critic conditioning. While the first-stage diffusion model is fully functional, the second-stage conditional model is still under training. This chapter provides an accurate overview of what has been achieved so far, along with a realistic roadmap for future development.

5.2 Summary of Completed Work

The work completed in this thesis can be divided into two phases.

5.2.1 Phase–1: Unconditional Diffusion Model (Completed)

- Implemented a complete DDPM forward process and DDIM reverse sampler.

- Trained a denoising model capable of reconstructing a 4-node truss geometry from noise.
- Demonstrated stable loss convergence, accurate reconstruction at all noise levels, and realistic generative trajectories.
- Verified that the model can generate the base truss directly from pure Gaussian noise.

This phase establishes a working diffusion backbone and validates the core generative modelling framework.

5.2.2 Phase–2 Preparation: FEM Pipeline and Dataset (Completed)

- Built a full FEM solver for small 2D truss structures.
- Evaluated the base truss and verified correctness of stress and displacement computations.
- Generated a dataset of **500 safe trusses** (random areas) using FEM-based safety filtering.
- Generated **5 perturbed safe geometries** using random node perturbations + FEM validation.
- Defined a compact 19-dimensional encoding that includes:
  - node coordinates,
  - member areas,
  - load vector,
  - one-hot load location.

These components form the foundation for conditional generative modelling.

5.2.3 Phase–2: Conditional Models (Partially Completed)

- Implemented a safety critic (classification + stress ratio regression).
- Implemented a conditional DDPM model architecture.
- Integrated sinusoidal time embeddings and conditioning embeddings.
- Started training the critic and diffusion model on the FEM-generated dataset.

At present, the training process is ongoing, and full conditional generations have not yet been produced.

5.3 Revised Discussion of Limitations

Based on the current stage of the project, the following limitations accurately reflect the implemented system.

5.3.1 Generative Capability Limited to One Base Geometry

The Stage–1 diffusion model learns only a single fixed geometry. It does not yet:

- generate diverse trusses,
- support varying member areas,
- incorporate structural constraints.

This is expected, as Stage–1 serves mainly as a proof-of-concept for the diffusion pipeline.

5.3.2 Conditional Generation Not Yet Completed

The Stage–2 system is implemented but not fully trained. Consequently:

- critic-guided sampling has not yet been demonstrated,
- safe trusses under arbitrary loads are not yet generated,
- full comparisons to optimization methods are not yet available.

5.3.3 Simplified Physics and Small-Scale Geometry

The FEM solver currently supports:

- small planar trusses,
- linear elasticity,
- static loading.

Real-world applications require richer physics and larger structures.

5.4 Future Research Directions

The following directions represent realistic and high-impact extensions.

5.4.1 Finish Training the Conditional Diffusion Model

Immediate next steps include:

- completing training on the merged dataset,
- validating critic accuracy,
- performing critic-guided sampling,

- generating safe truss designs conditioned on load.

This will complete the main objective of the thesis.

5.4.2 Generate and Evaluate Diverse Trusses

Once training is complete:

- generate hundreds of designs under different loads,
- evaluate them using FEM,
- compute safety, stiffness, and compliance metrics,
- compare against classical topology optimization.

5.4.3 Extend Encoding for Variable Topology

A major future direction is:

- adding discrete connectivity prediction,
- supporting varying number of nodes/members,
- using graph neural diffusion processes.

5.4.4 Enhance Physical Fidelity

Planned upgrades include:

- nonlinear FEM,
- buckling checks,
- dynamic and multi-load scenarios,
- multiple materials.

5.5 Conclusion

This project successfully demonstrates the feasibility of using diffusion models for structural generative design. The first-stage diffusion model validates that noise-to-structure generation is achievable. The second phase establishes the full physics-aware pipeline—FEM evaluation, dataset generation, critic networks, and conditional encodings—and is currently being trained.

While full conditional generative results are still in progress, the groundwork laid in this thesis provides a clear path toward a load-conditioned, safety-aware generative truss design system. As diffusion models and differentiable physics continue to evolve, such hybrid methods have strong potential for transforming structural design workflows.

5.6 Chapter Summary

This chapter summarized the actual completed work, clarified partially completed components, and outlined realistic next steps. Additional derivations, extended FEM outputs, and code listings appear in the Appendix.